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## Unit - 1 \* Important Ques in Blockchain and its Applications

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Q. What is Decentralization? Explain with the help of supply chain management example.

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Q. Discuss SHA 256 Algorithm with processing.

Cat-1  
Que.

Q. What is Hash pointer? Discuss temper detection using hash chain.

Q. Explain Merkle tree and structure of Merkle tree.

Cat-1  
Que.

Numerical

Q. Alice employs the RSA cryptosystem with the prime numbers  $P=7$  and  $Q=17$  to derive her public and private keys. Given that Alice's public key is 11, what will be her private key?

Cat-1  
Que.

Q. What is Digital Signature? Explain

Q. Explain Cryptocurrencies and its evolution.



## Unit - 2

Q Discuss DLTs and DApps.  
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Q What are Smart contracts. Explain various types of smart contracts.

cat-1 que

Q What is Blockchain. Draw and Explain Structure of Block and block headers.

Q. What are Bitcoin scripts? Explain with example.

cat-1 que

Q What are the steps for joining a bitcoin network? Explain.

cat-1 que

Q What is double spending? Explain how double spending is handled using blockchain

## Unit - 3

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Q Differentiate Permissioned and Permission less Blockchain.

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Q

Discuss { 1. Proof of Work (PoW)  
2. Proof of Stake (PoS)  
3. Proof of Burn.  
4. Proof of Elapsed Time

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Q What is Nonce? How is it used to solve puzzles in blockchain?



## Unit 3

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Q. Discuss Smart contracts over Closed Networks.

Q. What are the various Faults in a Distributed Systems? Explain.

Q. Explain Byzantine Faults and Byzantine Agreement Protocols.